ФЕДЕРАЛЬНОЕ АГЕНТСТВО

ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

РЕКОМЕНДАЦИИ ПО СТАНДАРТИЗАЦИИ

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КРИПТОГРАФИЧЕСКАЯ ЗАЩИТА ИНФОРМАЦИИ

Использование российских алгоритмов электронной подписи в протоколах и форматах сообщений на основе XML

Издание официальное

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

ljhgghich^ibkkb0/ ^hdmf_gkhb\kihevah\Zgjbk_kbckdbjbilh]jZnbq_kZdbjbjblfhih^ibkkbowrb
jh\Zgbby\uqbke_gabgyZq_g+b0y$&lhkdhevdhmf_glZjbhhij_^_e_g;b0/ ko_fug_y\eyxlky_fZg
lbq_kdbgZqbfufhbij_^_e_g'6yko_fb'6,* ko_fuhibkZgg\ug_Zklhysbjodhf_g^Zpbkyjbl\xlky
b^_glbqgufb hij_^_e_gbyf hibkZgguf \ > @

<jZa^_e_gZklhysbjodhf_g^Zpmbkl\ZgZ\eb\Zlxjlk[yh\Zgbdykh^_j`bfhfwme_f_glh\
hibku\Z_fuo \ jZa^_e_ b bo ij_^klZ\e_gbx

< mklZgZ\eb\Zlxjlk[yh\Zgbdykh^_j`bfhfwme_f_glb\]hij_^klZ\e_g^bexykha^Z
gbybijh_jdbwe_dljhgjh^cibkb0/ ^hdmf_gklZkihevah\Zgjbk_kbckdbkm^Zjkl_gguo
djbilh]jZnbq_kdbo Ze]hjblfh\ ih^ibkkb b owrbjh\Zgby

< mklZgZ\eb\Zlxjlk[yh\Zgbdykh^_j`bfhfwme_f_glb\]hij_^klZ\e_g^bexy\u
qbke_gby agZq_gby +0$& ;0/ ^hdmf_glZ k bkihevah\Zgb_f Ze]hjblfh\ hij_^

< mklZgZ\eb\Zlxjlk[yh\Zgbdykh^_j`bfhfwme_f_glb\]hij_^klZ\e_g^bexyi_j_
^Zqb dexq_\hc bgnhjfZpbb \ ih^ibku\Z_fhf ;0/ ^hdmf_gl_

5 Пространства имен XML и XML-префиксы

<gZklhysbjodhf_g^Zpbkyichevamx;Dk/ywe_f_gbuaq_luj_jcZaebqg;uOo/ ko_fdZ`
^hcbadhlhjukohhl_lkl\mm^gjhjhkljZgklbhf_g;0/ ljbkha^Zgjbk'ibkb;0/ ^hdmf_gkZb
\uqbke_gabgyZq_g+b0y$&\ khhl_lkl\kbgZklhysbjfbdhf_g^Zpb^yhfb`_gkihevah\Zlvky
ke_^mxsb^c_glbnbdZlhgjh\ghjhkljZgklbZ_g;0/ lhgd\Zdhgpij_^eh`_gbyy\ey_lky
qZklvbx^_glbnbdZlhrjnZietf:params:xml:ns:cpxmlsec HklZevgiujhkljZgklbZ_gy\
eyxlky \g_rgbfb b bo b^_glbnbdZlhju mdZau\Zxlky ^hihegbl_evgh \ aZ]heh

ljbf_qZg_b² ljhkljZgkl\bhf_g;0/ h^ghagZkghhl_lkl\mm^gjhc0/ ko_f_lZdbh[jZahf
mihlj_[e_gbl_dklgZa\Zg;b0y/ ko_fub^_glbnbpbjlmZdfkhhl_lkl\mxsjhkljZgklbhf_gGZa\Z
gby ;0/ ko_f ij_^klZ\e_gu \ lZ[ebp_

LZ[ebpZ² ;0/ ko_fu b^_glbnbdZlhju ijhkljZgkl\bhf_gb;DK/u

GZa\Zgb_ ;0/ ko_fu	B^_glbnbdZlhj ijhkljZgkl\Z bf_g	;0/ lj_nbdk
'6 ko_fZ	KWWS ZZZ Z RUJ [POGVLJ GV	
'6,* ko_fZ	KWWS ZZZ Z RUJ [POGVLJ GVLJ	
;6 ko_fZ	KWWS ZZZ Z RUJ ;0/6FKHPD [V	
&3;0/6(& ko_fZ	XUQ LHWI SDUDPV [POQV FS	[POVHFS[POVHF

ljbf_qZg_b² <gZklhysbjodhf_g^Zpbb6ykoko_fZ\ey_lk\kihfh]Zl_evgbhjobf_gy_lk^ey
hibkZgbiyj_^_e_gbba^jm]b;0/ ko_flhwlhfhmij_^_e_gwbey_f_glh6\ ko_fu^Zggujodhf_g^Zpbyo
g_ ijb\h^ylky

Qlh[ubkdexqbilvj_k_q_ghb^ygh bf_ggwuo_f_glbajZaebqgiujbkljZgkb\f_g;0/ \
gZklhysbjodhf_g^Zpb;0y/owe_f_gbuajZaebqg;uOo/ ko_fhf_q_gkuihfhsvxij_nbdkh\
hij_^_e_ggulZ[ebp_DZ`^ubcaij_^eh`_ggulZ[ebp_ij_nbdkh\eyk\h_]ihjhkljZgkl\Z
bf_g\u[jZgijhba\hevbgfhh`_l[ulvaZf_g_bggufh\k_owe_f_glZ_cabaf_g_gkbyfZglbq_
kdhc agZqbfhklb ;0/ ^hdmf_glZ

MdZaZigjbhkljZgkb\f_g;0/ bij_nbdkh\;0/ ^hdmf_ggl^he`gbf_lvjZaju\hkljhd
bijh[_eh\

ljbf_jaZ]heh\&d3Z;0/6(& ko_fueybkihevah\Zkghbiy_^_e_gbyhkdZevg;uOo/ ko_f
ijb_^_ggufb \ jZa^_e_ gZklhysbo j_dhf_g^Zpbc

```
<xs:schema
  xmlns:cpxmlsec="urn:ietf:params:xml:ns:cpxmlsec"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns:dsig11="http://www.w3.org/2009/xmldsig11#"
  targetNamespace="urn:ietf:params:xml:ns:cpxmlsec"
    elementFormDefault="qualified"
  version="0.4">
```


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6 Определение XML-схемы для элемента Signature XML-документа

We_f_ds:Signature y\ey_lky^gbbadhjg_\uwce_f_gl;0/ ^hdmf_gkZh^_j`bkle_ ^mxsmx bgnhjfZpbx

ijbnhjfbjh\Zgbbb^ibkb;0/ ^hdmf_gaZZq_gb^ibkbbgnhjfZpbx[Ze]hjblfZbo bo iZjZf_ljZo g_h[oh^bfuo ^ey nhjfbjh\Zgby we_dljhgghc ih^ibkb ;0/ ^hdmf_gl

ijb\uqbke_gabgbZq_g+0y$&;0/ ^hdmf_gbZnhjfZpbx[bkihevam_Ze]hjblf_u qbke_gby +0$& b agZq_gb_ +0$&

We_f_ds:Signature kh^_j`bl ke_^mxsb_ ^hq_jgb_ we_f_glu

We_f_ds:SignedInfo kf >Zgguwce_f_gkh^_j`bbgnhjfZpbx[bkihevam_fuo Ze]hjblfZbo ojm]bioZjZf_ljZg_h[oh^bf^ueoynhjfbjh\Zgby^ibkb;0/ ^hdmf_gkZbagZq_ gby +0$&

We_f_ds:SignatureValue kf >Zgguwce_f_gkh^_j`balgZq_gwbe_dljhgghc ibkb ;0/ ^hdmf_glZ beb agZq_gb_ +0$&

We_f_ds:KeyInfo kf >Zgguwce_f_gkh^_j`bbgnhjfZpbxdexqjh_jdbh^ ibkbb_]hagZq_gbe_bgnhjfZpbx-kbff_ljbqgbdexqbkihevam_ifb\uqbke_gabgbZq_gby +0$&

We_f_ds:Object. >Zgguwce_f_gh`_lkh^_j`Zbvgnhjfb^xeydhlhjhnchjfbjm _lky agZq_gb_ ih^ibkb beb \uqbkey_lky agZq_gb_ +0$&

We_f_ds:Signature hibku\Z_lky ke_^mxsbf hij_^_e_gb_f ;0/ ko_fu

```
<xs:element name="Signature" type="ds:SignatureType"/>

<xs:complexType name="SignatureType">
  <xs:sequence>
    <xs:element ref="ds:SignedInfo"/>
    <xs:element ref="ds:SignatureValue"/>
    <xs:element ref="ds:KeyInfo" minOccurs="0"/>
    <xs:element ref="ds:Object" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute name="Id" type="ID" use="optional"/>
</xs:complexType>
```

Ihegh_ hibkZgb_ \k_o ^hq_jgbo we_f_glh\ we_f_glZ GV 6LJQDWXUH ij_^klZ

6.1 Элемент SignedInfo

We_f_ds:SignedInfo y\ey_lky^hq_jgbwe_f_glhwe_f_gdsZSignature bkh^_j`bl k_[bgnhjfZpbx[bkihevam_Ze]hjblfZbo ojm]bioZjZf_ljZg_h[oh^bf^ueoynhjfbjh\Zgby ih^ibkb;0/ ^hdmf_gbZbagZq_gby +0$& We_f_ds:SignedInfo kh^_j`bke_^mxsb_ ^hq_jgb_ we_f_glu

We_f_ds:SignatureMethod kf >Zgguwce_f_gkh^_j`bbgnhjfZpbx[bkihev am_fhf Ze]hjblf_ ih^ibkb beb Ze]hjblf_ +0$&

We_f_ds:Reference kf >Zgguwce_f_gbkihevam^ekymdZaZgZ^yZgguk dhlhjufb \uihegyxlky ij_h[jZah\Zgby

We_f_ds:CanonicalizationMethod >Zgguwce_f_gkh^_j`bbgnhjfZpbx[bkihevam _fhfZe]hjblfjb_^_gbbdZghgbq_k\dh^mZggugoZoh^ysb\wke_f_gds:SignedInfo

We_f_ds:SignedInfo hibku\Z_lky ke_^mxsbf hij_^_e_gb_f ;0/ ko_fu

```
<xs:element name="SignedInfo" type="ds:SignedInfoType"/>

<xs:complexType name="SignedInfoType">
  <xs:sequence>
    <xs:element ref="ds:CanonicalizationMethod"/>
    <xs:element ref="ds:SignatureMethod"/>
    <xs:element ref="ds:Reference" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute name="Id" type="ID" use="optional"/>
</xs:complexType>
```

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Ihegh_ hibkZgb_ we_f_glZ ds:SignedInfo ij_^klZ\e_gh \ > @

6.1.1 Элемент SignatureMethod

We_f_gl ds:SignatureMethod y\ey_lk^yhq_jgbwe_f_glhwe_f_gl ds:SignedInfo b hij_^_ey_Ze]hjblfbkihevam_fuey^kha^Zgbyijh_jdbih^ibkb;0/ ^hdmf_gbZb Ze]hjblfbkihevam_fuey\uqbke_gbyZq_g+b0y$ & > eywlh]hZljb[mAlgorithm we_f_glZ ds:SignatureMethod ihf_sZ_lky b^_glbnbdZlhj khhl_lkl\mxsbc bkihevam_fhfm

B^_glbnbdZlhjuk jhkkbckdbo djbilh]jZnbq_kdbo Ze]hjblfh\ \uqbke_gby +0 bo \dexq_gby \ we_f_gl ds:SignatureMethod hibkZgu \

B^_glbnbdZlhju jhkkbckdbo djbilh]jZnbq_kdbo Ze]hjblfh\ \uqbke_gby +0

We_f_gl ds:SignatureMethod hibku\Z_lky ke_^mxsbf hij_^_e_gb_f ;0/ ko_fu

```
<xs:element name="SignatureMethod" type="ds:SignatureMethodType"/>

<xs:complexType name="SignatureMethodType" mixed="true">
  <xs:sequence>
    <xs:element name="HMACOutputLength" minOccurs="0"
            type="ds:HMACOutputLengthType"/>
    <xs:any namespace="##other" minOccurs="0" maxOccurs="unbounded"/>
    <!-- (0,unbounded) elements from (1,1) external namespace -->
  </xs:sequence>
  <xs:attribute name="Algorithm" type="anyURI" use="required"/>
</xs:complexType>
```

Ihegh_ hibkZgb_ we_f_gl ds:SignatureMethod ij_^klZ\e_gh \ > @

6.1.2 Элемент Reference

We_f_gl ds:Reference y\ey_lk^yhq_jgbwe_f_glhwe_f_gl ds:SignedInfo bfh`_l kh^_j`ZZlijb[mlu URI bType hibku\Zxsb^Zggu_dhkhjufbuihegyxlkiy_jZpbobwrb jh\Zgbby^jm]bij_h[jZah\ZgWe_f_gl ds:Reference ekh^_j`bkle_^mxs^bh_q_jgbwe_f_glu

We_f_gl ds:Transforms > Zgguwce_f_gl kh^_j`bkibkhij_h[jZah\Zg^bZgguomdZaZg guo \ Zljb[mlZo we_f_gl ds:Reference

We_f_gl ds:DigestMethod kf > Zgguwce_f_gl bkihevam^ekymdZaZgbkiyhev am_fh]h Ze]hjblfZ owrbjh\Zgby ^Zgguo mdZaZgguo ds:Reference[mlZo we_f_glZ

We_f_gl ds:DigestValue kf > Zgguwce_f_gl kh^_j`balgZq_gobw_r dhhZ^Zg guo mdZaZgguo \ Zljb[mlZo ds:Reference we_f_glZ

We_f_gl ds:Reference hibku\Z_lky ke_^mxsbf hij_^_e_gb_f ;0/ ko_fu

```
<xs:element name="Reference" type="ds:ReferenceType"/>

<xs:complexType name="ReferenceType">
  <xs:sequence>
    <xs:element ref="ds:Transforms" minOccurs="0"/>
    <xs:element ref="ds:DigestMethod"/>
    <xs:element ref="ds:DigestValue"/>
  </xs:sequence>
  <xs:attribute name="Id" type="ID" use="optional"/>
  <xs:attribute name="URI" type="anyURI" use="optional"/>
  <xs:attribute name="Type" type="anyURI" use="optional"/>
</xs:complexType>
```

Ihegh_ hibkZgb_ we_f_gl ds:Reference ij_^klZ\e_gh \ > @

We_f_gl 'LJHVW0HWKRG

We_f_gl ds:DigestMethod y\ey_lk^yhq_jgbwe_f_glhwe_f_gl ds:Reference bbf__l Zljb[mAlgorithm kh^_j`Zsbc^_glbnbdZZlhj Ze]hjblfbZwrbjh\Zg^bZyggu omdZaZg\gZulijb [mlZo we_f_gl ds:Reference

B^_glbnbdZlhjuk jhkkbckdbo djbilh]jZnbq_kdbo Ze]hjblfh\ \uqbke_gby \ we_f_gl ds:DigestMethod hibkZgu \

We_f_gl ds:DigestMethod hibku\Z_lky ke_^mxsbf hij_^_e_gb_f ;0/ ko_fu

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```
<xs:element name="DigestMethod" type="ds:DigestMethodType"/>

<xs:complexType name="DigestMethodType" mixed="true">
  <xs:sequence>
    <xs:any namespace="##other" processContents="lax"
        minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute name="Algorithm" type="anyURI" use="required"/>
</xs:complexType>
```

Ihegh_ hibkZgb_ wds:DigestMethod ij_^klZ\e_gh \ > @
We_f_gl 'LJHVW9DOXH

We_f_gds:DigestValue y\ey_lk^hq_jgbwfe_f_glbwfe_f_gdsZReference b^he`_g kh^_j`Zlv agZq_gbw_r dh^jZ_^klZ\e_gbwk_h_hl_lkl\bb AgZq_gbw_r dh^jZe`_hulv \uqbke_gh hl ^Zgguo mdZaZgguo \ ZdsbReferenceWe_f_glbwkZgghh \

We_f_gds:DigestValue hibku\Z_lky ke_^mxsbf hij_^_e_gb_f ;0/ ko_fu

```
<xs:element name="DigestValue" type="ds:DigestValueType"/>

<xs:simpleType name="DigestValueType">
  <xs:restriction base="base64Binary"/>
</xs:simpleType>
```

6.2 Элемент SignatureValue

We_f_gds:SignatureValue y\ey_lk^hq_jgbwfe_f_glbwfe_f_gdsZSignature b^he `_g kh^_j`Zlv agZq_gb_ we_dljhgghc ih^ibkb ;0/ ^hdmf_glZ beb agZq_gb_ +0$&

Kh^_j`bfhw_e_f_gdsZSignatureValue ijbbkihevah\Zjhkkbckdlbilh]jZnbq_kdbo Ze]hjblfh\ ih^ibkb b owrbjh\Zgby hibkZgh \

Kh^_j`bfhw_e_f_gdsZSignatureValue ijbbkihevah\Zjhkkbckdlbilh]jZnbq_kdbo Ze]hjblfh\ \uqbke_gby +0$& hibkZgh \

We_f_gds:SignatureValue hibku\Z_lky ke_^mxsbf hij_^_e_gb_f ;0/ ko_fu

```
<xs:element name="SignatureValue" type="ds:SignatureValueType" />

<xs:complexType name="SignatureValueType">
  <xs:simpleContent>
    <xs:extension base="base64Binary">
      <xs:attribute name="Id" type="ID" use="optional"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>
```

Ihegh_ hibkZgb_ wds:SignatureValue ij_^klZ\e_gh \ > @

6.3 Элемент KeyInfo

We_f_gds:KeyInfo y\ey_lk^hq_jgbwfe_f_glbwfe_f_gdsZSignature biha\heyi_l j_^Zlv ;0/ ^hdmf_gblg_nhjfZpbhxdexqijh_jdbh^ibkb _]hagZq_gbe_bbgnhjfZpbhx kbff_ljbqghf dexq_ bkihevam_fhf ijb \uqbke_gbb agZq_gby +0$&

<kemqZkeb;0/ ^hdmf_gjl_[m_lkyj_^Zldvexqijh_jdbh^ibkbg_h[oh^bbfkihev ah\Zlv h^bg ba ke_^mxsbo ^hq_jgbo wds:KeyInfo\ we_f_glZ

We_f_gds:KeyValue kf >Zgguwe_f_gjha\heyi_lj_^ZlavgZq_gbe_xqiZh _jdb ih^ibkb b _]h iZjZf_lju

We_f_gds:RetrievalMethod kf >Zgguwe_f_gjha\heyi_lj_^Zlvkuedgnz jZkiheh`_gb_ dexqZ ijh_jdb ih^ibkb _keb dexq gZoh^blky \g_ ih^ibku\Z_fh]h

We_f_gds:X509Data kf >Zgguwe_f_gjha\heyi_lj_^Zlv k_jlbnbdZl kf J ² ih^jZa^_e kh^_j`Zsbc dexq ijh_jdb ih^ibkb

We_f_gdsig11:DEREncodedKeyValue kf >Zgguwe_f_gjha\heyi_lj_^Zlv agZq_gb_ dexqZ ijh_jdb ih^ibkb b _]h iZjZf_lju
```

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```

l j b f _ q Z g _ b² W e _ f _ g d s:KeyValue b d s i g 1 1 : D E R E n c o d e d K e y V a l u e i h a \ h e y i _ l j _ ^ Z l v ^ g m b l m ` _ b g n h j f Z p b g h k b k i h e v a h \ Z g j Z _ a f e b q g i u h o ^ o h ^ h d \ _ _ i j _ ^ k l Z \ e _ g h b x k i h e v a h \ Z w g e _ f _ g l Z d s:KeyValue i h a \ h e y i _ l j _ ^ Z \ Z l a v g Z q _ g d b e _ x q Z h \ _ j d b h ^ i b k b _] h i Z j Z f _ l j u \ h l ^ _ e v g u b q _ j g b o w e _ f _ g l Z b k i h e v a h \ Z w g e _ f _ g l d s i g 1 1 : D E R E n c o d e d K e y V a l u e i h a \ h e y i _ l j _ ^ Z \ Z l a v g Z q _ g d b e _ x q Z i j h \ _ j d b i h ^ i b k b b _] h i Z j Z f _ l j u \ k h k l S u b j e c t P u b l i c K e y I n f o h i b k Z g g h c \ J

< k e m q _ Z _ k e b ; 0 / ^ h d m f _ g j l _ [m _ l k y j _ ^ Z l b v g n h j f Z p b h x k b f f _ l j b q g h d e x q _ b k i h e v a m _ f h j f b \ u q b k e _ g a b g b Z q _ g + b 0 y $ & g _ h [o h ^ b f k i h e v a h \ w Z e v _ f _ g d s:RetrievalMethod y \ e y x s b c k ^ y h q _ j g b w f e _ f _ g l b w f e _ f _ g d s Z K e y I n f o H i b k Z g b w e _ f _ g d s Z R e t r i e v a l M e t h o d i j _ ^ k l Z \ e _ g h \

l j b f _ q Z g _ b² G Z k l h y s b j _ d h f _ g ^ Z p m b k l Z g Z \ e b \ Z h x a f h ` g h k l v j _ ^ Z q b \ _ ^ _ g h c j Z k i h e h ` _ g b k b f f _ l j b q g h d e x q Z b k i h e v a m _ f h j f b \ u q b k e _ g a b g b Z q _ g + b 0 y $ & \ w e _ f _ g d s:RetrievalMethod l j b w l h f i j h p _ ^ m j \ u j Z [h l d b m d Z a Z g g h e] x q Z _] h j Z k i j _ ^ _ e _ g \ b y o h ^ y a Z h [e Z k l ^ v _ c k l \ b ^ y Z g g u o j _ d h f _ g ^ Z p b c

W e _ f _ g d s:KeyInfo h i b k u \ Z _ l k y k e _ ^ m x s b f h i j _ ^ _ e _ g b _ f ; 0 / k o _ f u

```
<xs:element name="KeyInfo" type="ds:KeyInfoType"/>

<xs:complexType name="KeyInfoType" mixed="true">
  <xs:choice maxOccurs="unbounded">
    <xs:element ref="ds:KeyName"/>
    <xs:element ref="ds:KeyValue"/>
    <xs:element ref="ds:RetrievalMethod"/>
    <xs:element ref="ds:X509Data"/>
    <xs:element ref="ds:PGPData"/>
    <xs:element ref="ds:SPKIData"/>
    <xs:element ref="ds:MgmtData"/>
    <!-- <xs:element ref="dsig11:DEREncodedKeyValue"/> -->
    <!-- DEREncodedKeyValue (XMLDsig 1.1) will use the any element -->
    <xs:any processContents="lax" namespace="##other"/>
    <!-- (1,1) elements from (0,unbounded) namespaces -->
  </xs:choice>
  <xs:attribute name="Id" type="ID" use="optional"/>
</xs:complexType>
```

l h e g h _ h i b k Z g b _ w e _ f _ g l Z d s i g 1 1 : K e y I n f o Z \ e _ g h \ > @

6.3.1 Элемент KeyValue

W e _ f _ g d s:KeyValue y \ e y _ l k ^ y h q _ j g b w f e _ f _ g l b w f e _ f _ g d s Z K e y I n f o b i h a \ h e y i _ l j _ ^ Z l v d e x q i j h \ _ j d b i h ^ i b k b \ i h ^ i b k u \ Z _ f h f ; 0 / ^ h d m f _ g l _

G Z k l h y s b j _ d h f _ g ^ Z p m b k l Z g Z \ e b \ Z e x l ^ m x s b h i h e g b l _ e v ^ g h u q _ j g b w e _ f _ g l w e _ f _ g l Z d s:KeyValue h i b k Z g g u _ \

w e _ f _ c p x m l s e c:GOSTR34102012-256-KeyValue
w e _ f _ c p x m l s e c:GOSTR34102012-512-KeyValue
w e _ f _ c p x m l s e c:GOSTR34102001KeyValue

W e _ f _ g d s:KeyValue h i b k u \ Z _ l k y k e _ ^ m x s b f h i j _ ^ _ e _ g b _ f ; 0 / k o _ f u

```
<xs:element name="KeyValue" type="ds:KeyValueType" />

<xs:complexType name="KeyValueType" mixed="true">
  <xs:choice>
    <xs:element ref="ds:DSAKeyValue"/>
    <xs:element ref="ds:RSAKeyValue"/>
    <!-- <xs:element ref="cpxmlsec:GOSTR34102012-256-KeyValue "/>
    <xs:element ref="cpxmlsec:GOSTR34102012-512-KeyValue "/>
    <xs:element ref="cpxmlsec:GOSTR34102001KeyValue "/> -->
        <!-- cpxmlsec:GOSTR34102012-256-KeyValue,
        cpxmlsec:GOSTR34102012-512-KeyValue,
        cpxmlsec:GOSTR34102001KeyValue will use the any element -->
    <xs:any namespace="##other" processContents="lax"/>
  </xs:choice>
</xs:complexType>
```

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lhegh_ hibkZgb_ we_f_glZ ds:KeyValueZ\e_gh \ > @
We_f_glu*2675 .H\9DOXH*2675 .H\9DOXH
*2675 .H\9DOXH
We_f_glu cpxmlsec:GOSTR34102012-256-KeyValue cpxmlsec:GOSTR34102012-512-KeyValue b cpxmlsec:GOSTR34102001KeyValue y\eyxlky^hq_jgbfw we_f_glZ fw we_f_glZ ds:KeyValue bf_xllb icpxmlsec:GOSTKeyValueType bdZ`^ u cagb k hklhb la^\m^hq_jgbo we_f_glh\
we_f_gl cpxmlsec:NamedCurve kh^_j`Zsbc b^_glbnbdZlhj weebilbq_kdhc djb\hc
we_f_gl cpxmlsec:PublicKey kh^_j`Zsbc agZq_gb_ dexqZ ijh_jdb ih^ibkb
>Zgguwe_f_gl iubgZ^e_`ZZhkljZgklbrh_gkij_nbdkl cpxmlsec b^_glbnbdZ dlhj lhjh]lbijZ\bezZdexq_g\b&3;0/6(& ko_fimbkZg\ujZa^_e_We_f_gl cpxmlsec:Named - Curve bcpxmlsec:PublicKey bf_xllb iudsig11:NamedCurveType bdsig11:ECPointType khhl\_lkl\_ggh hij_^_e_gby dhlhjuo ijbgZ^e_`Zl d '6,* ko_f_b hibkZgu \ > @
Lj_[h\Zgbyd kh^_j`bfhfmwe_f_glh\ cpxmlsec:GOSTR34102012-256-KeyValue cpxmlsec:GOSTR34102012-512-KeyValue cpxmlsec:GOSTR34102001KeyValue bbo^hq_jgb we_f_glh\ hibkZgu \ ²
We_f_gl cpxmlsec:GOSTR34102012-256-KeyValue cpxmlsec:GOSTR34102012-512-KeyValue и cpxmlsec:GOSTR34102001KeyValue hibku\Zxlkye_^mxsbhfij_^_e_gb_f &3;0/6(& ko_fu

```
<xs:element name="GOSTR34102012-256-KeyValue"
        type="cpxmlsec:GOSTKeyValueType" />

<xs:element name="GOSTR34102012-512-KeyValue"
        type="cpxmlsec:GOSTKeyValueType" />

<xs:element name="GOSTR34102001KeyValue"
        type="cpxmlsec:GOSTKeyValueType" />

<xs:complexType name="GOSTKeyValueType">
  <xs:sequence>
   <xs:element name="NamedCurve"
          type="dsig11:NamedCurveType" />
   <xs:element name="PublicKey"
          type="dsig11:ECPointType" />
  </xs:sequence>
</xs:complexType>
```

6.3.2 Элемент RetrievalMethod

We_f_gl ds:RetrievalMethod y\ey_lk^hq_jgb we_f_glh we_f_gl ds:KeyInfo biha\h ey_^Zlvkkuedm Zdexq_\mbgnhjfZpbdkhlhjZ\ZkiheZ]Z_lfkykl_hlebqgbhfih^ibku\Z_ fh]h ;0/ ^hdmf_glZ >ey wlh]h bkihevamkURIyType Zljb[mlu
We_f_gl ds:RetrievalMethod bf__l^hq_jgb we_f_gl ds:Transforms bkihevam_^fey ij_h[jZah\Zgby ^Zgguo gZ dhlhju_ mdZau\Zxlds:RetrievalMethod we_f_glZ
We_f_gl ds:RetrievalMethod hibku\Z_lky ke_^mxsbf hij_^_e_gb_f ;0/ ko_fu

```
<xs:element name="RetrievalMethod" type="ds:RetrievalMethodType" />

<xs:complexType name="RetrievalMethodType">
  <xs:sequence>
   <xs:element ref="ds:Transforms" minOccurs="0" />
  </xs:sequence>
  <xs:attribute name="URI" type="anyURI" />
  <xs:attribute name="Type" type="anyURI" use="optional" />
</xs:complexType>
```

lhegh_ hibkZgb_ we_f_glh\ ds:RetrievalMethod ds:Transforms ij_^klZ\e_gh \ > @

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6.3.3 Элемент X509Data

We_f_gl ds:X509Data y\ey_lky\hq_jgbwfe_f_glwfe_f_gl ds:KeyInfo bih a\heyi_lj_ ^Zlvk_^_gby k_jlbnbdZjlb_ihfhsbdhlhjhlbehknhjfbjh\Zgbq_gwbe__dljhgghc ih^ibkb ;0/ ^hdmf_glZ eb[h kZf k_jlbnbdZl k dexqhf ijh_jdb ih^ibkb

We_f_gl ds:X509Data hibku\Z_lky ke_^mxsbf hij_^_e_gb_f ;0/ ko_fu

```
<xs:element name="X509Data" type="ds:X509DataType"/>

<xs:complexType name="X509DataType">
  <xs:sequence maxOccurs="unbounded">
    <xs:choice>
      <xs:element          name="X509IssuerSerial"
                           type="ds:X509IssuerSerialType"/>
      <xs:element name="X509SKI" type="base64Binary"/>
      <xs:element name="X509SubjectName" type="string"/>
           <xs:element name="X509Certificate" type="base64Binary"/>
      <xs:element name="X509CRL" type="base64Binary"/>
      <!-- < xs:element ref="dsig11:X509Digest"/> -->
           <!-- The X509Digest element (XMLDSig 1.1) will use the
           any element -->
      <xs:any namespace="##other" processContents="lax"/>
    </xs:choice>
  </xs:sequence>
</xs:complexType>
```

Ihegh_ hibkZgb_ we_f_glZ ds:X509DataZ\e_gh \ > @

6.3.4 Элемент DEREncodedKeyValue

We_f_gl dsig11:DEREncodedKeyValue y\ey_lkyZkrbj_gb_hfij_^_e_g;0/ ko_fu we_ f_glZ ds:KeyInfo hibkZgglh]h bih a\heyi_lj_^Zlvexqjh_jdbh^ibkbLj_[h\Zgbky h ^_j`bfhfm we_f_gl dsig11:DEREncodedKeyValue hibkZgu \

We_f_gl dsig11:DEREncodedKeyValue hibku\Z_lky ke_^mxsbf hij_^_e_gb_f ;0/ k

```
<!-- targetNamespace=»http://www.w3.org/2009/xmldsig11#» -->

<xs:element name="DEREncodedKeyValue"
        type="dsig11:DEREncodedKeyValueType" />
<xs:complexType name="DEREncodedKeyValueType">
  <xs:simpleContent>
    <xs:extension base="base64Binary">
      <xs:attribute name="Id" type="ID" use="optional"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>
```

Ihegh_ hibkZgb_ wdsig11:DEREncodedKeyValue ij_^klZ\e_gh \ > @

7 Использование российских криптографических алгоритмов

<^Zgghjf Za^_eg_Zklhysbj_odhf_g^Zmbkl ZgZ\eb\Zlxjl_kyh\Zgbdykh^_j`bfhfwme_ f_glh\hibku\Z_fuiZa^_e_b_]hij_^klZ\e_gibkbkihevah\Zjgbkkbckdbkm^Zjkl_g guo djbilh]jZnbq_kdbo Ze]hjblfh\ ih^ibkb owrbjh\Zgby b \uqbke_gby +0$&

7.1 Использование российских криптографических алгоритмов при создании подписи XML-документов

7.1.1 Использование алгоритма хэширования в элементе DigestMethod

Bkihevah\ZZgb]hjblfoZwrbjh\Zg=bHyKLJ ² k^ebghocwr dh^Z [bl\ we_f_gl_ 'LJHVW0HWKRG

>eyZe]hjblfoZwrbjh\Zgbyij_^_e_gg\h=hHKLJ ² k^ebghocwr dh^Z [bl ^he`_gbkihevah\Zlvkey_^mxsbkc^_glbnbdZlurn:ietf:params:xml:ns:cpxmlsec: algorithms:gostr34112012-256

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l j b f _ q Z g _ b² < g Z k l h y s b _ o d h f _ g ^ Z p ^ b e y o k _ b ^ _ g l b n b d Z l h j k k b c k d b o b o i l h] j Z n b q _ k d b o
Z e] h j b l f h \ i _ j _ g h k k l j h d b \ g m l j b a g Z q _ g b y b ^ _ g l b n b d Z l h j Z b l h q d Z \ _] h d h g p _ g _ m q b l

l j b f _ j w e _ f _ g l d Z ds:DigestMethod ^ e y Z e] h j b l f Z o w r b j h \ Z g b y i j _ ^ _ e _ g g h \ ] h
= H K L J ² k ^ e b g h c o w r d h ^ Z [b l

<ds:DigestMethod Algorithm=
“urn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr34112012-256” />

A g Z q _ g b _ [b l g h] h w r d h ^ i Z _ ^ k l Z \ e _ g g \ h] b ^ _ i h k e _ ^ h \ Z l _ e v g b h a k l b [Z c l \ i h j y ^
d _ O L W W O b l \ h l Q ^ G b l j D \ Q l _ V H h i b k Z g g \ h > @ h e ` g h u l v i h f _ s _ g h k h ^ _ j ` b f h w e _ f _ g l Z
ds:DigestValue h i b k Z g g h] h \
B k i h e v a h Z Z e g b j _ b l f o Z w r b j h \ Z g = b y K L J ² k ^ e b g h o c w r d h ^ Z [b l \
w e _ f _ g l _ ' L J H V W 0 H W K R G
> e y Z e] h j b l f o Z w r b j h \ Z g b y i j _ ^ _ e _ g g \ h] = H K L J ² k ^ e b g h o c w r d h ^ Z [b l
^ h e ` _ g b k i h e v a h \ Z l v k e y _ ^ m x s b c ^ _ g l b n b d Z l u r n : i e t f : p a r a m s : x m l n s : c p x m l s e c :
a l g o r i t h m s : g o s t r 3 4 1 1 2 0 1 2 - 5 1 2
l j b f _ j w e _ f _ g l d Z ds:DigestMethod ^ e y Z e] h j b l f Z o w r b j h \ Z g b y i j _ ^ _ e _ g g h \ ] h
= H K L J ² k ^ e b g h c o w r d h ^ Z [b l

<ds:DigestMethod Algorithm=
“urn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr34112012-512” />

A g Z q _ g b _ [b l g h] h w r d h ^ i Z _ ^ k l Z \ e _ g g \ h] b ^ _ i h k e _ ^ h \ Z l _ e v g b h a k l b [Z c l \ i h
j y ^ d _ O L W W O b l \ h l Q ^ G b l j D \ Q l _ V H ^ h e ` g h u l v i h f _ s _ g h k h ^ _ j ` b f h w e _ f _ g l Z ds:Digest -
Value
B k i h e v a h \ Z e g b j _ b l f Z o w r b j h \ Z g b = y K L J ² j Z a ^ _ e \ w e _ f _ g l _
' L J H V W 0 H W K R G
> e y h [_ k i _ q _ g \ b y a f h ` g h k b b k i h e v a h \ Z l g b y e] h \ j _ f _ g g h Z j h o b \ g h] h j Z g _ g b y 0 /
^ h d m f _ g l i h \ ^ i b k Z g g i u j b o i h f h s b Z e] h j b l f Z i j _ ^ _ e _ g g \ h = H K L J ² j Z a ^ _ e Z l Z d
` _ i j h \ _ j d b w e _ d l j h g i h ^ c i b k b w e _ f _ g ds:DigestMethod g _ h [o h ^ b i f h f _ s Z l v k e _ ^ m x s b c
b ^ _ g l b n b d Z u r n : i e t f : p a r a m s : x m l n s : c p x m l s e c : a l g o r i t h m s : g o s t r 3 4 1 1
W e _ f _ g ds:DigestMethod f h ` _ l k h ^ _ j ` Z l ^ v h q _ j g b w e _ f _ g cpxmlsec:NamedParame -
ters h i j _ ^ _ e y x s b c i Z j Z f _ l j u ^ e y Z e] h j b l f Z o w r b j h \ Z g b y
W e _ f _ g cpxmlsec:NamedParameters b f _ _ Z l j b [n b URI k h ^ _ j ` Z s b i Z j Z f _ l j u Z e] h j b l f Z
o w r b j h \ Z g b Z y j Z f _ l j u a Z ^ Z \ Z _ f b u ^ _ _ g l b n b d Z l h j Z Z f b j _ d h f _ g ^ m _ d l k ^ y b j h \ Z l k h h l \ _ l
k l \ b b k > @ h i m k d Z _ b l k i h e v a h \ Z b g ^ b _ g l b n b d Z l h Z j h \ i Z j Z f _ l j h h i j _ ^ _ e _ g g u > o i h ^ j Z a
^ _ e @
? k e b w e _ f _ g cpxmlsec:NamedParameters h l k m l k l \ m i j b e h ` _ g b ^ h e ` g h b k i h e v a h \ Z l v
i Z j Z f _ l j u k h h l \ _ l k l \ m x s b ^ _ g l b n b d Z l h y Z f h j Z i Z j Z f _ l j h i d-GostR3411-94-Crypto -
ProParamSet h i j _ ^ _ e _ g g h f m \ > @
W e _ f _ g cpxmlsec:NamedParameters h i b k u \ Z _ l k y k e _ ^ m x s b f h i j _ ^ _ e _ g b _ f ; 0 / k o _ f u

<xs:element name=”NamedParameters”
type=”cpxmlsec:NamedParametersType” />

l j b f _ j w e _ f _ g l d Z ds:DigestMethod ^ e y Z e] h j b l f Z o w r b j h \ Z g b y i j _ ^ _ e _ g g h \ ] h
= H K L J ² j Z a ^ _ e

<ds:DigestMethod Algorithm=
“urn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr3411”>
<!-- id-GostR3411-94-CryptoProParamSet -->
<cpxmlsec:NamedParameters URI=”urn:oid:1.2.643.2.2.30.1” />
</ds:DigestMethod>

A g Z q _ g b _ [b l g h] h w r d h ^ i Z _ ^ k l Z \ e _ g g \ h] b ^ _ i h k e _ ^ h \ Z l _ e v g b h a k l b [Z c l \ i h
j y ^ d _ O L W W O b l \ h l Q ^ G b l j D \ Q l _ V H ^ h e ` g h u l v i h f _ s _ g h k h ^ _ j ` b f h w e _ f _ g l Z ds:Digest -
Value

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7.1.2 Использование алгоритма подписи в элементе SignatureMethod

Bkihevah\Zegbj_blfZ^ibkb=HKLJ ² k^ebghcexqZ [bl\we_f_gl_ 6LJQDWXUH0HWKRG

>eyZe]hjblfZ^ibkbhij_^_e_gg\h=HKLJ ² kdexqhif^ibkb [bl^he`_gbkihevah\Zkvek_y^mxsbc^_glbnbdZurn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr34102012-gostr34112012-256

ljbf_j we_f_gdsZSignatureMethod kmdZaZgbZe]hjblfZ^ibkbhij_^_e_gg\h]h=HKL J ² k dexqhf ih^ibkb [bl

<ds:SignatureMethod Algorithm=
“urn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr34102012-
gostr34112012-256” />

AgZq_gbih^ibkbknhjfbjh\Zggh_khhl_lkl\bkb Ze]hjblfhfhij_^_e_ggu\f=HKLJ ² kdexqhfh^ibkb [bl^he`_gh[ulvij_^klZ\e_gkhkhhl_lkl\bkbJ \dh^bjh\d_ EDVH bihf_sdsSignatureValue_gl

Bkihevah\Zegbj_blfZ^ibkb=HKLJ ² k^ebghcexqZ [bl\we_f_gl_ 6LJQDWXUH0HWKRG

>eyZe]hjblfZ^ibkbhij_^_e_gg\h=HKLJ ² kdexqhif^ibkb [bl^he`_gbkihevah\Zkvek_y^mxsbc^_glbnbdZurn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr34102012-gostr34112012-512

ljbf_j we_f_gldsZSignatureMethod kmdZaZgbZe]hjblfZhij_^_e_gg\h]h=HKL J ² k dexqhf ih^ibkb [bl

<ds:SignatureMethod Algorithm=
“urn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr34102012-
gostr34112012-512” />

AgZq_gbih^ibkbknhjfbjh\Zggh_khhl_lkl\bkb Ze]hjblfhfhij_^_e_ggu\f=HKLJ ² kdexqhfh^ibkb [bl^he`_gh[ulvij_^klZ\e_gkhkhhl_lkl\bkbJ \dh^bjh\d_ EDVH bihf_sdsSignatureValue_gl

Bkihevah\Zke]hjblfZ^ibkb=HKLJ ² ih^jZa^_e \we_f_gl_ 6LJQDWXUH0HWKRG

>ey h[_ki_q_gbyafh`ghklbkihevah\Zgbyhe]h\j_f_ggh]hjob\gh]hjZg_gby;0/ ^hdmf_glh\^ibkZggujb ihfhsbZe]hjblfZhij_^_e_ggh\]h=HKLJ ² ih^jZa^_e ZlZd`_ijh_jdbwe_dljhggh^ibkb we_f_gldsSignatureMethod g_h[oh^bfhihf_sZlvke_^mxsbc^_glbnbdZurn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr34102001-gostr3411

ljbf_j we_f_gldsZSignatureMethod kmdZaZgbZe]hjblfZhij_^_e_ggh\]h=HKL J ² ih^jZa^_e

<ds:SignatureMethod Algorithm=
“urn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr34102001-gostr3411” />

AgZq_gbih^ibkbknhjfbjh\Zggh_khhl_lkl\bkb Ze]hjblfhfhij_^_e_ggu\f=HKLJ ² ih^jZa^_e kdexqhif^ibkb [bl^he`_gh[ulvij_^klZ\e_gkhkhhl_lkl\bbk J \dh^bjh\d_ EDVH bihfdssSignatureValue_gl

7.2 Использование российских криптографических алгоритмов при вычислении значения HMAC в XML-документе

:e]hjblfowrbjh\Zgbhij_^_e_gg\u=HKLJ ² fh`_l[ulvbkihevah\Zgq_kl_Ze]hjblfZ \uqbke_gby +0$& \khhl_lkl\bbk > imgdl @ bJ ²

7.2.1 Использование хэширования ГОСТ Р 34.11—2012 с длиной хэш-кода 256 бит в элементе SignatureMethod

>eyZe]hjblfoZwrbjh\Zgbhij_^_e_gg\h=HKLJ ² k^ebghcwr dh^Z [bl^he`_gkihevah\Zkvek_y^mxsbc^_glbnbdZurn:ietf:params:xml:ns:cpxmlsec:algorithms:hmac-gostr34112012-256

ljbf_jwe_f_gldsZSignatureMethod kaZ^ZggZe]hjblfhofwrbjh\Zgbhij_^_e_gguf\ =HKL J ² k ^ebghc owr dh^Z [bl

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<ds:SignatureMethod Algorithm=
“urn:ietf:params:xml:ns:cpxmlsec:algorithms:hmac-gostr34112012-256” />

J_amevjZ[hluZe]hjblfHMAC_GOSTR3411_2012_256 hij_^_e_gg\hJh ² imgdl ^he`_fulvij_^klZ\e_gb^_ihke_^h\Zl_evgbhakl[Zcl\ ihjy^dOLWWOH HQGLDQ dh^bjh\d_ EDVH b ihf_s_g ds:SignatureValue hibkZgguc \

7.2.2 Использование алгоритма хэширования ГОСТ Р 34.11—2012 с длиной хэш-кода 512 бит в элементе SignatureMethod

>eyZe]hjblfoZwrbjh\Zgbiyj_^_e_gg\h=HKL ² k^ebghcowr dh^Z[bl^he `_gbkihevah\Zlkky^mxsbc^_glbnbdZlurn:ietf:params:xml:ns:cpxmlsec:algo rithms:hmac-gostr34112012-512

ljbf_j we_f_glZds:SignatureMethod k aZ^ZggufZe]hjblfhf owrbjh\Zgby =HKL J ² k^ebghc owr dh^Z [bl

<ds:SignatureMethod Algorithm=
“urn:ietf:params:xml:ns:cpxmlsec:algorithms:hmac-gostr34112012-512” />

J_amevjZ[hluZe]hjblfHMAC_GOSTR3411_2012_512 hij_^_e_gg\hJh ² imgdl ^he`_fulvij_^klZ\e_gb^_ihke_^h\Zl_evgbhakl[Zcl\ ihjy^dOLWWOH HQGLDQ dh^bjh\d_ EDVH b ihf_s_g \ we_f_gl ds:SignatureValue

7.3 Передача информации о ключевом материале

>ey i_j_^Zqbbgnhjf Zpbbh dexq_\hffZl_jbZe_Ze]hjblfh\hij_^_e_ggu\o =HKLJ ² b\=HKLJ ² ih^jZa^_e fh]m[ulvbkihevah\Zgxu_ih^ oh^ysbkihkh[uhij_^_e_gg\u> @GZklhysbj_dhf_g^Zp^bbihegbl_ehgjh_^_eyxklih kh[i_j_^Zqbexq_cjh_jdbih^ibkbZe]hjblfZHKLJ ² \we_f_glsig11: DEREncodedKeyValue hibkZgg\hf b\^hq_jgbwe_f_glvZe_f_glsZKeyValue hibkZg gh]h \

7.3.1 Передача ключа проверки подписи в элементе DEREncodedKeyValue

GZklhysbj_dhf_g^Zpmbklf Zg Z\eb\Zhxalfh`ghkliv_j_^Zqbexq Zjh\_jdbih^ib kb khhl\_lkl\mxsZ_e]hjblfmHKLJ ² k^ebghc [blbeb [bl\we_f_gl_ dsig11:DEREncodedKeyValue hibkZgghf GZklhysbj_dhf_g^Zpbbd`_mklZgZ\eb\Z xl\hafh`ghkilvj_^Zqbexq Zjh_jdbih^ibkbkhhl_lkl\mxsZ_e]hjblfmij_^_e_gghfm =HKLJ ² ih^jZa^_e \dsig11:DEREncodedKeyValue hibkZgghf \

>eykha^Zgbwe_dljhggihc^ibkbkhhl_lkl\mxsZ_e]hjblfmHKLJ ² b ijh_jdbwe_dljhggihc^ibkbkhhl_lkl\mxsZ_e]hjblfmij_^_e_gghfmHKLJ ² ih^jZa^_e dexqijh_jdbih^ibkbb_]hiZjZf_ljui_j_^Z\Z_fu_we_f_glsig11: DEREncodedKeyValue ^he`gu[ulv ihf_s_gu\ kljmdlmSubjectPublicKeyInfo kf J b aZdh^bjh\Zgu \ khhl_lkl\bb k J

ljbf_j we_f_glZdsig11:DEREncodedKeyValue

<dsig11:DEREncodedKeyValue>
<!-- значение ключа проверки подписи -->
</dsig11:DEREncodedKeyValue>

7.3.2 Передача ключа проверки подписи алгоритма ГОСТ Р 34.10—2012 с длиной ключа 256 бит в элементе GOSTR34102012-256-KeyValue

GZklhysbj_dhf_g^Zpmbklf Zg Z\eb\Zhxalfh`ghkliv_j_^Zqbbgnhjf Zpbbhf_klhgZoh` ^_gbbexq Zjh_jdbih^ibkbkhhl_lkl\mxsZ_e]hjblfmHKLJ ² k^ebghc [bl >eyi_j_^Zqbbgnhjf Zpbbhf_klhgZoh`^_gbexq\ZZljb[mType we_f_glds:Reference beb ds:RetrievalMethod hibkZgg\uo b khhl_lkl_gbh_sZ_lkkye_^mxsbc^_glbnb dZlhurn:ietf:params:xml:ns:cpxmlsec:types:gostr34102012-256-keyvalue

>eyi_j_^Zqbexq Zjh_jdbih^ibkbih^ibku\Z_fh0/ ^hdmf_gl_h[oh^bifjh_^klZ \blvdexq nhjfZl_hibkZgg^hfyihey subjectPublicKey kljmdlmSubjectPublicKeyInfo kf J gh[_aij_^klZ\e_gb2y&7(7675,1*bdh^bjh\Zgb(y5 >Zgghij_^klZ\ e_gb^he`gfulvaZdh^bjh\ZgEHDVHbihf_s_gh ^hq_jgbwe_f_glpxmlsec:PublicKey we_f_glZpxmlsec:GOSTR34102012-256-KeyValue ZgZeh]bqwgh_f_glds:RSAKeyValue hibkZgghfm @hij_^_e_gh0/ ko_fu^eywe_f_glpxmlsec:GOSTR34102012-256-Key - Value lcpxmlsec:PublicKey hibkZgh \

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> e y i _ j _ ^ Z q b b ^ _ g l b n b d Z l w j e Z b i l b q _ k d j b b c h c i Z j Z f _ l j h d l e x q Z j h \ _ j d b b h ^ i b k b \ i h ^ i b k u \ Z _ f h , 0 / ^ h d m f _ g b l k _ i h e v a m _ Z l l k y [m URI w e _ f _ g d p Z xmlsec:NamedCurve h i b k Z g g h \ ] h l Z j Z f _ l j u a Z ^ Z \ Z _ f b u ^ _ _ g l b n b d Z l h j Z , f b j _ d h f _ g ^ m _ d l k ^ y b j h \ Z l k h h l \ _ l k l \ b k b > @ Z k l h y s j b _ d h f _ g ^ Z p b b m k d Z b x k l i h e v a h \ Z b g ^ b _ g l b n b d Z l h Z j , h \ Z j Z f _ l j h \ h i j _ ^ _ e _ g g u o \ J

l j b f _ j w e _ f _ g d p Z xmlsec:GOSTR34102012-256-KeyValue

```
<cpxmlsec:GOSTR34102012-256-KeyValue>
  <!-- id-GostR3410-2001-CryptoPro-A-ParamSet -->
  <cpxmlsec:NamedCurve URI="urn:oid:1.2.643.2.2.35.1" />
  <cpxmlsec:PublicKey>
        <!-- значение ключа проверки подписи -->
  </cpxmlsec:PublicKey>
</cpxmlsec:GOSTR34102012-256-KeyValue>
```

7.3.3 Передача ключа проверки подписи алгоритма ГОСТ Р 34.10—2012 с длиной ключа 512 бит в элементе GOSTR34102012-512-KeyValue

G Z k l h y s j b _ d h f _ g ^ Z p b b k l Z g Z \ e b \ Z h x a f h ` g h k l v j _ ^ Z q b b g n h j f Z p b b f _ k l h g Z o h ` ^ _ g b b l e x q Z j h \ _ j d b b h ^ i b k b k h h l \ _ l k l \ m x s Z _ e] h j b l f = n H K L J ² k ^ e b g h c [b l > e y i _ j _ ^ Z q b b g n h j f Z p b b f _ k l h g Z o h ` ^ _ g e b x b q \ Z Z l j b [m Type w e _ f _ g d s \ Reference b e b ds:RetrievalMethod h i b k Z g g \ u o b k h h l \ _ l k l \ _ g g h _ s Z _ l k k y e _ ^ m x s b b ^ c _ g l b n b d Z l h j urn:ietf:params:xml:ns:cpxmlsec:types:gostr34102012-512-keyvalue

> e y i _ j _ ^ Z q b e x q Z j h \ _ j d b b h ^ i b k b b i h ^ i b k u \ Z _ f h , 0 / ^ h d m f _ g l _ h [o h ^ b i f j h _ ^ k l Z \ b l v d e x q n h j f Z l _ h i b k Z g g ^ e y i h e y subjectPublicKey k l j m d l m SubjectPublicKeyInfo k f J g h [_ a i j _ ^ k l Z \ e _ g \ b 2 y & 7 (7 6 7 5 , 1 b d h ^ b j h \ Z g b (y 5 > Z g g h i j _ ^ k l Z \ e _ g b ^ _ h e ` g [h u l v a Z d h ^ b j h \ Z g h E D V H b i h f _ s _ g h ^ h q _ j g b w e _ f _ g d p xmlsec:PublicKey w e _ f _ g b p Z xmlsec:GOSTR34102012-512-KeyValue Z g Z e h] b q w g h _ f _ g d s m RSAKeyValue h i b k Z g g h \ f m @ h i j _ ^ _ e _ g b 0 / k o _ f u ^ e y w e _ f _ g l d p xmlsec:GOSTR34102012-512-Key - Value b c p xmlsec:PublicKey h i b k Z g h \

> e y i _ j _ ^ Z q b b ^ _ g l b n b d Z l w j e Z b i l b q _ k d j b b c h c i Z j Z f _ l j h d l e x q Z j h \ _ j d b b h ^ i b k b \ i h ^ i b k u \ Z _ f h , 0 / ^ h d m f _ g b l k _ i h e v a m _ Z l l k y [m URI w e _ f _ g d p Z xmlsec:NamedCurve h i b k Z g g h \ ] h l Z j Z f _ l j u a Z ^ Z \ Z _ f b u ^ _ _ g l b n b d Z l h j Z , f b j _ d h f _ g ^ m _ d l k ^ y b j h \ Z l k h h l \ _ l k l \ b k b > @ Z k l h y s j b _ d h f _ g ^ Z p b b m k d Z b x k l i h e v a h \ Z b g ^ b _ g l b n b d Z l h Z j , h \ Z j Z f _ l j h \ h i j _ ^ _ e _ g g u o \ J

l j b f _ j w e _ f _ g d p Z xmlsec:GOSTR34102012-512-KeyValue

```
<cpxmlsec:GOSTR34102012-512-KeyValue>
  <!-- id-tc26-gost-3410-12-512-paramSetA -->
  <cpxmlsec:NamedCurve URI="urn:oid:1.2.643.7.1.2.1.2.1" />
  <cpxmlsec:PublicKey>
        <!-- значение ключа проверки подписи -->
  </cpxmlsec:PublicKey>
</cpxmlsec:GOSTR34102012-512-KeyValue>
```

7.3.4 Передача ключа проверки подписи алгоритма ГОСТ Р 34.10—2001, подраздел 6.2, в элементе GOSTR34102001KeyValue

> e y h [_ k i _ q _ g \ b n y a f h ` g h k b b k i h e v a h \ Z b g b y e ] h \ j _ f _ g g h Z j h o b \ g h j b h j Z g _ g b , 0 / ^ h d m f _ g l h h \ ^ i b k Z g g i u j o i h f h s b Z e ] h j b l f Z h i j _ ^ _ e _ g g h ] = n H K L J ² i h ^ j Z a ^ _ e Z l Z d ` i j h \ _ j d b w e _ d l j h g i h ^ c i b k b Z l j b [m URI w e _ f _ g d s \ RetrievalMethod b e b ds:Reference ^ e y i _ j _ ^ Z q b b g n h j f Z p b b f _ k l h g Z o h ` ^ _ g b e x q Z j h \ _ j d b i h ^ i b k b g _ h [o h ^ b f h i h f _ s Z l v k e _ ^ m x s b b ^ _ g l b n b d Z l h j urn:ietf:params:xml:ns:cpxmlsec: - types:gostr34102001-keyvalue

> e y i _ j _ ^ Z q b e x q Z j h \ _ j d b b h ^ i b k b k h h l \ _ l k l \ m x s Z _ e] h j b l f = n H K L J ² i h ^ j Z a ^ _ e g _ h [o h ^ b i f j h _ ^ k l Z \ b l v d e x q n h j f Z l _ h i b k Z g g ^ e y i h e y subjectPublicKey k l j m d l m j u SubjectPublicKeyInfo ^ e y GostR3410-2012-256-PublicKey k f J g h [_ a i j _ ^ k l Z \ e _ g \ b 2 y & 7 (7 6 7 5 , 1 b d h ^ b j h \ Z g b (y 5 > Z g g h i j _ ^ k l Z \ e _ g ^ b _ e ` g [h u l v a Z d h ^ b j h \ Z g h E D V H b i h f _ s _ g h ^ h q _ j g b w e _ f _ g d p xmlsec:PublicKey w e _ f _ g d p Z xmlsec:GOSTR -

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34102001KeyValue Z g Z e h] b w g h _ f _ g d s m RSAKeyValue h i b k Z g g h \ f m @ h i j _ ^ _ e _ g g u o _ /
k o _ f u ^ e y w e _ f _ g l d p x mlsec:GOSTR34102001KeyValue b cpxmlsec:PublicKey h i b k Z g \ h

> e y i _ j _ ^ Z q b o ^ _ g l b n b d Z l w j e Z b i l b q _ k d j b c h c i Z j Z f _ l j h d l e x q Z j h \ _ j d b h ^ i b k b \
i h ^ i b k u \ Z _ f h f / ^ h d m f _ g b l k _ i h e v a m _ Z l k j y [m URI w e _ f _ g d p Z xmlsec:NamedCurve h i b
k Z g g h \ ] h l Z j Z f _ l j u a Z ^ Z \ Z _ f b u ^ _ g l b n b d Z l h j Z f b j _ d h f _ g ^ m _ d l k ^ y b j h \ Z l k h h l
\ _ l k l \ b k b > @ a Z k l h y s b _ d h f _ g ^ Z p h b m k d Z b x k l i h e v a h \ Z b g ^ b _ g l b n b d Z l h j h \ Z j Z f _ l j h \
h i j _ ^ _ e _ g g u o \ > i h ^ j Z a ^ _ e @
l j b f _ j w e _ f _ g p x Z mlsec:GOSTR34102001KeyValue

```
<cpxmlsec:GOSTR34102001KeyValue>
  <!-- id-GostR3410-2001-CryptoPro-A-ParamSet -->
  <cpxmlsec:NamedCurve URI="urn:oid:1.2.643.2.2.35.1" />
  <cpxmlsec:PublicKey>
        <!-- значение ключа проверки подписи -->
  </cpxmlsec:PublicKey>
</cpxmlsec:GOSTR34102001KeyValue>
```

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Приложение А
(справочное)

Полная XML-схема


```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Declare helper entities to avoid overrunning right margin of text while importing
schemata.-->
<!DOCTYPE schema [
  <!ENTITY xmldsiguri
    "http://www.w3.org/TR/2008/REC-xmldsig-core-20080610">
]>

<xs:schema
  xmlns:cpxmlsec="urn:ietf:params:xml:ns:cpxmlsec"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns:dsig11="http://www.w3.org/2009/xmldsig11#"
  targetNamespace="urn:ietf:params:xml:ns:cpxmlsec"
    elementFormDefault="qualified"
  version="0.4">

  <xs:import namespace="http://www.w3.org/2000/09/xmldsig#" />

  <xs:import namespace="http://www.w3.org/2009/xmldsig11#" />

  <xs:element name="NamedParameters"
          type="cpxmlsec:NamedParametersType" />

  <xs:complexType name="NamedParametersType">
    <xs:attribute name="URI" type="xs:anyURI" use="required" />
  </xs:complexType>

  <xs:complexType name="GOSTKeyValueType">
    <xs:sequence>
      <xs:element name="NamedCurve"
              type="dsig11:NamedCurveType" />
      <xs:element name="PublicKey" type="dsig11:ECPointType" />
    </xs:sequence>
  </xs:complexType>

  <xs:element name="GOSTR34102012-256-KeyValue"
          type="cpxmlsec:GOSTKeyValueType" />
  <xs:element name="GOSTR34102012-512-KeyValue"
          type="cpxmlsec:GOSTKeyValueType" />
  <xs:element name="GOSTR34102001KeyValue"
          type="cpxmlsec:GOSTKeyValueType" />

</xs:schema>
```


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Приложение Б
(справочное)

Примеры

< ^Zgghifjbeh`_glkh^_j`Zlhykgh\guij_bf_jubkihevah\Zjhkykbckdzeo]hjblfhw\e_dljhgghc ih^ibkb ijb kha^Zgbb we_dljhgghc ih^ibkb ;0/ ^hdmf_glh\

Ijbf_qZg_b² < gZklhyslpodhf_g^Zpbky_oijb_^_ggu ih^ibkZggu 0/ ^hdmf_gbfu_xl klijmdlmij m^klZ\e_gg hm^xbl_evkdbf bq_jgbfwe_f_glZ fbeym^h[klqZ_gb;0/ ^hdmf_gbZZ^uc bawe_f_glbdjhf_i_j\]hbf__hlklmQbkehhlklmimh_ebqb\Zg_Zky^bgeydZ`^hjeh`_gghjhmq_j g_]hwe_f_gkZ^Zggu jo dhf_g^Zpbh^ybgbklklmijbgylkqblZJvZ\gufj_fijh[_eZfAgZq_gbyibkb kh^_j`Zsb_kiybf_jZoihemq_kgmq_lhwflhbkh[_gghkfbmbaf_gblvkkvebkeihh[_eh\\h^ghf hlklmi_ mf_gvrblv beb m_ebqblv

Б.1 Подпись XML-документа с использованием алгоритма, определенного в ГОСТ Р 34.10—2012, с длиной хэш-кода 256 бит в элементе DigestMethod

>eykha^Zgibhy^ibkb 0/ ^hdmf_gilZ^klZ\e_gg\h^Zggghifjbf_j_[uebkihevahkZ_glbnbdZl iheghklvx b^_glbqguc k_jlbnbdZlm \ J ² ijbeh`_gb_ :

Dhhj^bgZlZ *x* dexqZ ijh_jdb ih^ibkb k_jlbnbdZlZ jZ\gZ

0x971566CEDA436EE7678F7E07E84EBB7217406C0B4747AA8FD2AB1453C3D0DFBA

Dhhj^bgZlZ *y* jZ\gZ

0xAD58736965949F8E59830F8DE20FC6C0D177F6AB599874F1E2E24FF71F9CE643

Khhl_lkl\mxsbc dexq *d* hj^Zl b_kgb

0xBFCF1D623E5CDD3032A7C6EABB4A923C46E43D640FFEAAF2C3ED39A8FA399924

Qbkeh *k* jZ\gh

0x5782C53F110C596F9155D35EBD25A06A89C50391850A8FEFE33B0E270318857C

Qbkehj *Z* \gh

0x054D1DABB161D63424F8DABB2800708B00F78DA7582699E8F2F0A521C7CE8144

Ih^ibkZgguc ;0/ ^hdmf_gl \u]ey^bl ke_^mxsbf h[jZahf

```
<?xml version="1.0" encoding="utf-8"?>
<root>
  <DataToSign Id="ToSign">Data</DataToSign>
  <Signature xmlns="http://www.w3.org/2000/09/xmldsig#">
    <SignedInfo>
          <CanonicalizationMethod Algorithm="http://www.w3.org/TR/2001/REC-
          xml-c14n-20010315" />
          <SignatureMethod Algorithm="urn:ietf:params:xml:ns:
          cpxmlsec:algorithms:gostr34102012-gostr34112012-256" />
      <Reference URI="#ToSign">
        <Transforms>
                <Transform Algorithm="http://www.w3.org/TR/2001/REC-xml-
                c14n-20010315" />
        </Transforms>
        <        DigestMethod Algorithm="urn:ietf:params:xml:ns:cpxmlsec:algorithms:
                 gostr34112012-256" />
        <DigestValue>9QLsxPPo7LlX6IXqwzjcNDmbFuCCGivQ1s61hcPuITM=</DigestValue>
      </Reference>
    </SignedInfo>
    <        SignatureValue>jcQJhWtWbTCV7bjFky5vGXXUFigc74FXRi79lZnFHK7pMjpeiN2H+
             3xyQ4O//nzs1Ln/oqwzvu9zpaH3Q0BPaw==</SignatureValue>
    <KeyInfo>
      <KeyValue>
        <GOSTR34102012-256-KeyValue
          xmlns="urn:ietf:params:xml:ns:cpxmlsec">
          <NamedCurve URI="urn:oid:1.2.643.2.2.36.0" />
                <PublicKey>ut/Qw1MUq9KPqkdHC2xAF3K7TugHfo9n525D2s5mFZdD5pwf90/i4vF0m
                Fmr9nfRwMYP4o0Pg1mOn5RlaXNYrQ==</PublicKey>
```

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```
        </GOSTR34102012-256-KeyValue>
      </KeyValue>
    </KeyInfo>
  </Signature>
</root>
```

Ih^ibkZgguc ;0/ ^hdmf_gl \ dh^bjh\d_ EDVH \u]ey^bl ke_^mxsbf h[jZahf

77u/PD94bWwgdmVyc2lvbj0iMS4wIiBlbmNvZGluZz0idXRmLTgiPz48cm9vdD4NCiAgIDxEYXRhVG9TaWdu
IElkPSJUb1NpZ24iPkRhdGE8L0RhdGFUb1NpZ24+DQogICA8U2lnbmF0dXJlIHhtbG5zPSJodHRwOi8vd3d3
LnczLm9yZy8yMDAwLzA5L3htbGRzaWcjIj4NCiAgICAgIDxTaWduZWRJbmZvPg0KICAgICAgICAgPENhbm9u
aWNhbGl6YXRpb25NZXRob2QgQWxnb3JpdGhtPSJodHRwOi8vd3d3LnczLm9yZy9UUi8yMDAxL1JFQy14bWwt
YzE0bi0yMDAxMDMxNSIgLz4NCiAgICAgICAgIDxTaWduYXR1cmVNZXRob2QgQWxnb3JpdGhtPSJ1cm46aWV0
ZjpwYXJhbXM6eG1sOm5zOmNweG1sc2VjOmFsZ29yaXRobXM6Z29zdHIzNDEwMjAxMi1nb3N0cjM0MTEyMDEy
LTI1NiIgLz4NCiAgICAgICAgIDxSZWZlcmVuY2UgVVJJPSIjVG9TaWduIj4NCiAgICAgICAgICAgIDxUcmFu
c2Zvcm1zPg0KICAgICAgICAgICAgICAgPFRyYW5zZm9ybSBBbGdvcml0aG09Imh0dHA6Ly93d3cudzMub3Jn
L1RSLzIwMDEvUkVDLXhtbC1jMTRuLTIwMDEwMzE1IiAvPg0KICAgICAgICAgICAgPC9UcmFuc2Zvcm1zPg0K
ICAgICAgICAgICAgPERpZ2VzdE1ldGhvZCBBbGdvcml0aG09InVybjppZXRmOnBhcmFtczp4bWw6bnM6Y3B4
bWxzZWM6YWxnb3JpdGhtczpnb3N0cjM0MTEyMDEyLTI1NiIgLz4NCiAgICAgICAgICAgIDxEaWdlc3RWYWx1
ZT45UUxzeFBQbzdMbFg2SVhxd3pqY05EbWJGdUNDR2l2UTFzNjFoY1B1SVRNPTwvRGlnZXN0VmFsdWU+DQog
ICAgICAgICA8L1JlZmVyZW5jZT4NCiAgICAgIDwvU2lnbmVkSW5mbz4NCiAgICAgIDxTaWduYXR1cmVWYWx1
ZT5qY1FKaFd0V2JUQ1Y3YmpGa3k1dkdYWFVGaWdjNzRGWFJpNzlsWm5GSEs3cE1qcGVpTjJlKzN4eVE0Ty8v
bnpzMUxuL29xd3p2dTl6cGFIM1EwQlBhdz09PC9TaWduYXR1cmVWYWx1ZT4NCiAgICAgICAgLZXlJbmZvPg0K
ICAgICAgICAgPEtleVZhbHVlPg0KICAgICAgICAgICAgPEdPU1RSMzQxMDlwMTItMjU2LUtleVZhbHVlIHht
bG5zPSJ1cm46aWV0ZjpwYXJhbXM6eG1sOm5zOmNweG1sc2VjIj4NCiAgICAgICAgICAgICAgIDxOYW1lZEN1
cnZlIFVSST0idXJuOm9pZDoxLjIuNjQzLjIuMi4zNi4wIiAvPg0KICAgICAgICAgICAgPFB1YmxpY0tl
eT51dC9RdzFNVXE5S1Bxa2RIQzJ4QUYzSzdUdWdIZm85bjUyNUQyczVtRlpkRDVwd2Y5MC9pNHZGMG1GbXI5
bmZSd01ZUDRvMFBnMW1PbjVSbGFYTllyUT09PC9QdWJsaWNLZXk+DQogICAgICAgICAgICA8L0dPU1RSMzQx
MDIwMTItMjU2LUtleVZhbHVlPg0KICAgICAgICAgPC9LZXlWYWx1ZT4NCiAgICAgIDwvS2V5SW5mbz4NCiAg
IDwvU2lnbmF0dXJlPg0KPC9yb290Pg==
```

### Б.2 Подпись XML-документа с использованием алгоритма, определенного в ГОСТ Р 34.10—2012, с длиной хэш-кода 512 бит в элементе DigestMethod

>eykha^Zgibhy^ibkb0/ ^hdmf_gilZ^klZ\e_gg\h^]Zgghifjbf_j_[uebkihevahkZ_glbnbdZl
iheghklvx b^_glbqguc k_jlbnbdZlm \J ² ijbeh`_gb_ :

Dhhj^bgZlZ x dexqZ ijh\_jdb ih^ibkb k_jlbnbdZlZ jZ\gZ

0x07134627CE7FC6770953ABA4714B38AF8DE764B8870A502C2F4CC2D05541459A18DA3B9D4EBC09BC
06CB2EA1856A03747561CF04C34382111539230A550F1913

Dhhj^bgZlZ y jZ\gZ

0x7E08A434CB2FA300F8974E3FF69A4BCDF36B6308E1D7A56144693A35E11CBD14D502916E680E35FE1E
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Khhl\_lkl\mxsbc dexq d hj^Zi\b_kgb

0x3FC01CDCD4EC5F972EB482774C41E66DB7F380528DFE9E67992BA05AEE462435757530E641077CE587
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Qbkeh k jZ\gh

0x72ABB44536656BF1618CE10BF7EADD40582304A51EE4E2A25A0A32CB0E773ABB23B7D8FDD8FA5EEE91
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Qbkehjl Z\gh

0x33DEF8422879AA68482339BC65E5DCA9A5D77E80C5C0371DB13D3B88F4CCA8A89ED3CE85849231DD61
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Ih^ibkZgguc ;0/ ^hdmf_gl \u]ey^bl ke_^mxsbf h[jZahf

```
<?xml version="1.0" encoding="utf-8"?>
<root>
  <DataToSign Id="ToSign">Data</DataToSign>
  <Signature xmlns="http://www.w3.org/2000/09/xmldsig#">
    <SignedInfo>
      <        CanonicalizationMethod Algorithm="http://www.w3.org/TR/2001/REC-
               xml-c14n-20010315" />
      <        SignatureMethod
               Algorithm="urn:ietf:params:xml:ns:cpxmlsec:algorithms:
               gostr34102012-gostr34112012-512" />
```


ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ
```

```
    <Reference URI="#ToSign">
      <Transforms>
        <        Transform Algorithm="http://www.w3.org/TR/2001/REC-xml-
                 c14n-20010315" />
      </Transforms>
      <        DigestMethod
               Algorithm="urn:ietf:params:xml:ns:cpxmlsec:algorithms:gos
               tr34112012-512" />
      <        DigestValue>wiOFD9D7zKHNlo58t/9tUtCJA5ZO9vmDhMlt3HIkyXZvQxIp5PE+txwsIAV
               fUIOULvGTFxAZlwuHTB+qD5s54g==</DigestValue>
    </Reference>
  </SignedInfo>
  <     SignatureValue>dn+oWg6n3wJ20kBmO1GvURc4SuZ3h3nKXYWy4uHdmeS2nlTlNWFKca4fTBlc+f
        pnCS8IEVNFX25Ndh4UXJLLNl2/L0wtancFiA+xRYzFgzUGW+pWIfyfvBdsSspbweZyJUWajqN3lDR
        ZDchycEApNlqDpTtes8BpNrXSh+Cpg+c=</SignatureValue>
  <KeyInfo>
    <KeyValue>
      <GOSTR34102012-512-KeyValue
        xmlns="urn:ietf:params:xml:ns:cpxmlsec">
        <NamedCurve URI="urn:oid:1.2.643.7.1.2.1.2.2" />
        <        PublicKey>ExkPVQojORURgkPDBM9hdXQDaoWhLssGvAm8Tp072hiaRUFV0MJMLyxQCo
                 e4ZOeNrzhLcaSrUwl3xn/OJ0YTB/0PW2XgHNnjv8oca7EIUwbn2tRbqLtqHv41DmhukQ
                 LVFL0c4TU6aURhpdfhCGNr881LmvY/Tpf4AKMvyzSkCH4=</PublicKey>
      </GOSTR34102012-512-KeyValue>
    </KeyValue>
  </KeyInfo>
 </Signature>
</root>
```


Ih^ibkZgguc ;0/ ^hdmf_gl \ dh^bjh\d_ EDVH \u]ey^bl ke_^mxsbf h[jZahf

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```


**Б.3 Подпись XML-документа с использованием алгоритма, определенного в ГОСТ Р 34.10—2001, подраздел 6.1, в элементе SignatureMethod**

> e ykh a ^ Z gibny^ i b k b,0/ ^ h d m f _ gilj_Z ^ k l Z \ e _ g g\h^]Z g g hifj b f _ j _ [ u e b k i h e v a h k_Zgl b n b d Z l i h e g h k l v x b ^ _ g l b q g u c k _ j l b n b d Z l m \ > i h ^ j Z a ^ _ e @

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ
```

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Khhl_lkl\mxsbc dexq d hj^Zl\b_kgb
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Qbkeh k jZ\gh
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Qbkehjl\Z\gh
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Ih^ibkZgguc ;0/ ^hdmf_gl \u]ey^bl ke_^mxsbf h[jZahf


```
<?xml version="1.0" encoding="utf-8"?>
<root>
  <DataToSign Id="ToSign">Data</DataToSign>
  <Signature xmlns="http://www.w3.org/2000/09/xmldsig#">
    <SignedInfo>
      <        CanonicalizationMethod
               Algorithm="http://www.w3.org/TR/2001/REC-xml-c14n-20010315" />
      <        SignatureMethod Algorithm="urn:ietf:params:xml:ns:cpxmlsec:algorithms:gos
               tr34102001-gostr3411" />
      <Reference URI="#ToSign">
        <Transforms>
          <            Transform Algorithm="http://www.w3.org/TR/2001/REC-xml-
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        </Transforms>
        <          DigestMethod Algorithm="urn:ietf:params:xml:ns:cpxmlsec:algorithms:gos
                   tr3411" />
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      </Reference>
    </SignedInfo>
    <      SignatureValue>n2UHtdu25fPzJNYyojbNTq52V1D3UBVQqI5xNhdYopDpMjpeiN2H+3xyQ4O//
           nzs1Ln/oqwzvu9zpaH3Q0BPaw==</SignatureValue>
    <KeyInfo>
      <KeyValue>
        <GOSTR34102001KeyValue xmlns="urn:ietf:params:xml:ns:cpxmlsec">
          <NamedCurve URI="urn:oid:1.2.643.2.2.36.0" />
                <PublicKey>hJVodWACGkB1CM0TjDGJLP3lBQN6Q1z0bSsP508yfleP68wWuZWIA9CafIW
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        </GOSTR34102001KeyValue>
      </KeyValue>
    </KeyInfo>
  </Signature>
</root>
```


Ih^ibkZgguc ;0/ ^hdmf_gl \ dh^bjh\d_ EDVH \u]ey^bl ke_^mxsbf h[jZahf

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Б.4 Подпись XML-документа с передачей X.509-сертификата в элементе KeyInfo

> e y k h a ^ Z g i b y ^ i b k b 0 / ^ h d m f _ g l j Z _ ^ k l Z \ e _ g g \ h ^ j Z g g h i f j b f _ j _ [u e b k i h e v a h k Z g l b n b d Z l
i h e g h k l v x b ^ _ g l b q g u c k _ j l b n b d Z l m \ J ² i j b e h ` _ g b _ :
D h h j ^ b g Z l Z x d e x q Z i j h \ _ j d b i h ^ i b k b k _ j l b n b d Z l Z j Z \ g Z
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D h h j ^ b g Z l Z y j Z \ g Z
0xAD58736965949F8E59830F8DE20FC6C0D177F6AB599874F1E2E24FF71F9CE643
K h h l \ _ l k l \ m x s b c d e x q d h j ^ Z l b _ k g b
0xBFCF1D623E5CDD3032A7C6EABB4A923C46E43D640FFEAAF2C3ED39A8FA399924
Q b k e h k j Z \ g h
0x5782C53F110C596F9155D35EBD25A06A89C50391850A8FEFE33B0E270318857C
Q b k e h j Z \ g h
0x054D1DABB161D63424F8DABB2800708B00F78DA7582699E8F2F0A521C7CE8144
I h ^ i b k Z g g u c ; 0 / ^ h d m f _ g l \ u] e y ^ b l k e _ ^ m x s b f h [j Z a h f

```
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<root>
  <DataToSign Id="ToSign">Data</DataToSign>
  <Signature xmlns="http://www.w3.org/2000/09/xmldsig#">
    <SignedInfo>
      <        CanonicalizationMethod Algorithm="http://www.w3.org/TR/2001/REC-xml-
               c14n-20010315" />
      <        SignatureMethod Algorithm="urn:ietf:params:xml:ns:cpxmlsec:algorithms:gos
               tr34102012-gostr34112012-256" />
      <Reference URI="#ToSign">
        <Transforms>
          <              Transform Algorithm="http://www.w3.org/TR/2001/REC-xml-
                         c14n-20010315" />
        </Transforms>
        <            DigestMethod Algorithm="urn:ietf:params:xml:ns:cpxmlsec:algorithms:gos
                     tr34112012-256" />
        <DigestValue>9QLsxPPo7LlX6IXqwzjcNDmbFuCCGivQ1s61hcPuITM=</DigestValue>
      </Reference>
    </SignedInfo>
    <      SignatureValue>jcQJhWtWbTCV7bjFky5vGXXUFigc74FXRi79lZnFHK7pMjpeiN2H+3xyQ4O//
           nzs1Ln/oqwzvu9zpaH3Q0BPaw==</SignatureValue>
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            <X509Certificate>MIICYjCCAg+gAwIBAgIBATAKBggqhQMHAQEDAjBWMSkwJwYJKoZIhvcN
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```

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

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Б.5 Подпись XML-документа с ключом проверки подписи ГОСТ Р 34.10—2012 256 бит в элементе DEREncodedKeyValue

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l h ^ i b k Z g g u c ; 0 / ^ h d m f _ g l \ u] e y ^ b l k e _ ^ m x s b f h [j Z a h f


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      <CanonicalizationMethod Algorithm="http://www.w3.org/TR/2001/REC-xml-c14n-20010315" />
      <SignatureMethod Algorithm="urn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr34102012-gostr34112012-256" />
      <Reference URI="#ToSign">
        <Transforms>
          <Transform Algorithm="http://www.w3.org/TR/2001/REC-xml-c14n-20010315" />
        </Transforms>
        <DigestMethod Algorithm="urn:ietf:params:xml:ns:cpxmlsec:algorithms:gostr34112012-256" />
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      </Reference>
    </SignedInfo>
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    <KeyInfo>
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      </DEREncodedKeyValue>
    </KeyInfo>
  </Signature>
</root>
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l h ^ i b k Z g g u c ; 0 / ^ h d m f _ g l \ d h ^ b j h \ d _ E D V H \ u] e y ^ b l k e _ ^ m x s b f h [j Z a h f


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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

Библиография

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Dhjj_dlhj *И.А. Королева*
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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ